

THE SMOOTH TRANSITION

**A STUDY ON THE IMPACT OF THE FRENCH BOARD GENDER
QUOTA ON FIRM VALUE**

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Bachelor Thesis

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The Smooth Transition: A Study on The Impact of the French Board Gender Quota on Firm Value

Abstract:

This study examines the impact of the French board gender quota on firm value, drawing comparisons with previous research. Utilizing methodology akin to Ahern and Dittmar (2012), who investigated the Norwegian board gender quota, we analyze the effects of the French quota enacted in 2011. We use the pre-quota cross-sectional variation in board gender diversity to instrument for exogenous changes to corporate boards resulting from the law. Contrary to prior findings on the impact of board gender quotas, our results reveal no evidence supporting the notion that the French quota negatively affects firm value. Despite the quota mandating an involuntary increase in female board members, we observe no significant impact on immediate stock return or long-term indicators such as Tobin's Q and return on equity. Additionally, board size and tenure remain unaffected by the quota, suggesting a smooth transition rather than forced adjustments. By comparing with the Norwegian experience, characterized by rushed compliance, we argue that the extended compliance period and softer sanctions upon failure to comply in France facilitated a gradual adjustment process, mitigating adverse effects on firm value. These results challenge prior assumptions and provide policymakers with insights on implementing gender quotas without compromising firm performance while advancing gender equality in corporate leadership.

Keywords:

Board Gender Quota, Copé Zimmerman law, Diversity, Firm Value, Board skills

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1 Introduction

Gender equality in the workplace has become a prominent political issue, resulting in the implementation of board gender quotas in several countries. Historically, women have been underrepresented in higher-level positions, commonly referred to as a “glass ceiling” (Bertrand, Black, Jensen, and Lleras-Muney, 2019). To address this issue, European countries have passed hard laws to increase board gender diversity, such as the ones in Norway, Italy, Spain, and France. Others, like Luxembourg, the UK, and Sweden, have encouraged a specific percentage of female board members through soft laws (Ferreira, Ginglinger, Paris-Dauphine, and Skalli, 2021). As stated by the directive passed by the European parliament in 2022, starting in July 2026, all listed companies in the European Union must have at least 40% of their non-executive directors and 33% of their executive directors of the underrepresented gender (European Commission, 2022). This new EU directive will require many firms to replace their current board members to meet the new requirements, making it crucial how a gender quota affects firms and their boards. There is, however, limited evidence on the effect of gender quotas on board and firm outcomes. A prominent study by Ahern and Dittmar (2012) on the Norwegian gender quota from 2003 found significant costs to shareholders and a decrease in board skills. As such, this paper employs a methodology akin to Ahern and Dittmar (2012) to examine the impact of the French gender quota imposed in 2011. The objective is to determine whether, irrespective of policy implementation details and contextual factors, gender quotas have adverse effects on board skills and firm value.

Our study addresses the following research question:

Are board gender quotas unfavorable for firm value?

In 2011, the Copé-Zimmerman law was enacted in France, mandating that 40% of board members must be of the underrepresented gender by 2017, with an interim target of 20% by 2014. This law applies to all firms headquartered in France that are publicly listed and non-listed firms with over 500 employees or revenues exceeding €50m over the previous three years (République Française, 2011). Any board appointment that violates the stipulated provisions or does not help remedy the non-compliance will be considered invalid. In addition, failure to comply with any of the mandated proportion requirements leads to a suspension of payment of director fees until the gender diversity in the board of directors meets the stipulated quota threshold.

We have used both the event study methodology and *instrumental variable* approach used by Ahern and Dittmar (2012) and compared our findings to their prominent study on the Norwegian gender quota. Thus, understanding the French context in relation to the Norwegian one is crucial. Publicly listed companies in Norway were initially encouraged in 2003 to voluntarily meet a gender quota threshold of 40% within one and a half years. However, when compliance fell short, companies were granted a two-year transition period to reach the same threshold, but this time, facing penalties if they did not comply. Upon no fulfillment in 2008, the company would be dissolved. This is in contrast to France, where firms had six years, and not two years as in Norway, to fulfill the requirement stipulated by the law. Additionally, the penalty for non-compliance in France, suspension of payment of director fees, is more lenient than in Norway, where firms risk being dissolved. Not only did policy implementation details differ between the two countries, but the female labor supply setting also differed, with France having a less favorable situation, as found by Adams and Kirchmaier (2015).

According to our research on the French example, the implementation of gender quotas does not, regardless of context and policy implementation details, have an adverse effect on firm value. The results of the regressions performed to investigate the effects of the French law showed no unfavorable results. Our analysis first reveals that the stock market anticipated a smooth transition, as there was no difference in industry-adjusted immediate returns for companies that were closer or further away from the quota threshold. Secondly, our long-term study on how firm value was affected also did not support the finding of any adverse effects. This is in contrast with the study performed by Ahern and Dittmar (2012), which found both a negative immediate stock market effect and unfavorable outcomes on long-term firm value following the law’s implementation. In Norway, it was found that the skill level of board members decreased as a result of the law, which subsequently caused the decline in firm value (Ahern and Dittmar, 2012). We attribute the lack of adverse firm value effects in France to the fact that our regression showed that the law had no negative effect on the percentage of board members who had industry-relevant backgrounds or strong financial skills. We argue that this was enabled by

the more extended transition period, which enabled a *gradual* replacement of new competent female board members replacing the incumbent male directors.

To conduct our analysis, we gathered data from a diverse set of sources. Data on our treatment group, companies located in France that were listed on Euronext Paris, along with our control group, US-situated companies listed on American exchanges, was mainly sourced from Refinitiv Eikon. For firms that had some missing data points, we acquired the remaining data using alternative sources. Missing financial data was obtained from CapitalIQ, and missing data on the board was gathered from the companies' annual reports. Our dataset included almost 100 French firms, a sample size comparable to that used by Ahern and Dittmar (2012).

Our research involved an event study, per the methodology of Ahern and Dittmar (2012), to investigate the impact of the quota announcement on companies' stock prices. We divided the French companies into two groups based on their female board participation at the time of the event study: Low diversity and High diversity. We chose a threshold value at the median of 8.33% in order to create two approximately equally large groups.

Since the law had been frequently debated in parliament prior to its enactment, it was expected that a well-informed market would have already factored in the changes before the date when the statute was passed. Instead, we examine the point in time when the bill was initially brought up in parliament, January 20th, 2010. On the same day, an attention-grabbing news article called "Shaking up the corporate world" (France 24, 2010) was published. We found no significant difference in the average industry-adjusted stock returns between companies with fewer or more female directors in France, as the difference of 0.83% was insignificant. This suggests that the stock market did not foresee any negative impact on companies with fewer female board members. Our study results are robust and consider various factors, including board size, firm size, and assets. We also had a control group consisting of American firms listed in the US.

In order to evaluate the long-term influence of the quota on firm value, an instrumental variable procedure was used. Similar to Ahern and Dittmar (2012), we utilized the pre-quota cross-sectional variation in female board representation to instrument for exogenous changes to corporate boards following the quota. As all public limited firms were required to have at least 20% female directors in 2014 and 40% in 2017, those companies that had a higher percentage of female directors prior to the quota were less restricted than the ones with lower female representation. The results from instrumental variable estimates reveal that the quota did not negatively impact Tobin's Q, as the point estimate -0.279% was insignificant. These results include firm and time-fixed effects to control for both observed and unobserved heterogeneity between the firms and yearly trends. Placebo tests run on US firms, as also done by Ahern and Dittmar (2012), show no significant connection between the increase of women on boards due to the quota and subsequent changes in firm value. These outcomes demonstrate that the quota mandated by the law and its subsequent board reorganizations did not have any unfavorable effect on firm value. Furthermore, a robustness check was conducted by examining how the law altered the firms' return on equity (ROE), and consistent with our prior findings, no significant effect was found.

To gain a deeper understanding of why there was no loss in value, we analyzed how women were added to the boards and how the quota affected the skills of directors. This is important as Ahern and Dittmar (2012) concluded that the negative effect on firm value resulting from the Norwegian gender quota was attributable to a decrease in board skills. In this study, we define board-specific skills as the percentage of board members who have either an industry-specific background or a strong financial background. Our instrumental variable regression on board skills showed a positive coefficient of 0.161%, which was insignificant, meaning that board skills cannot be concluded to have been negatively affected as a result of the law.

We conclude that our study did not find any negative effects of the gender quota as a result of board-specific skills not having declined, as they did in Norway (Ahern and Dittmar, 2012). This was probably made possible by the *gradual* replacement of board members in France, whereas in Norway, the negative effects were a result of firms making drastic adjustments to meet the deadline of the quota (Ahern and Dittmar, 2012). As the female supply setting was less preferable in France than in Norway (Adams and Kirchmaier, 2015), our study concludes that the smooth replacement was enabled by the longer transition period and possibly the more lenient penalty for non-compliance. These results may offer guidance to policymakers on maintaining firm value while promoting greater gender equality in the boardroom.

Related literature This paper contributes to the large and growing literature on board gender quotas’ impact on firm value (Ahern and Dittmar, 2012; Adams and Ferreira, 2009; Bernile, Bhagwat, and Yonker, 2018; Ginglinger and Raskopf, 2023; Likitapiwat, Treepongkaruna, Jiraporn, and Uyar, 2022; Greene, Intintoli, and Kahle, 2020; Burrow, Fedorets, and Gibert, 2018). More specifically, by studying the immediate stock price reaction at the announcement of the board gender quota and studying its long-term impact on Tobin’s Q in a Norwegian firm setting, Ahern and Dittmar (2012) argue that forced increased board gender diversity caused deterioration of firm value. They saw a significant negative long-term impact, as well as an adverse immediate stock market reaction. A similar study by Burrows et al. (2018) investigated the financial outcomes of the German law from 2015 that mandated a supervisory board (two-tier system) gender quota for approximately 100 large corporations. The authors found no significant effect on the financial outcomes of the firms. Also, Greene et al. (2020) studied the immediate stock market reaction in 2018 to the announcement of the first mandated board gender quota in the US. In California, firms were allowed a transition period of a bit over one year to have at least one female on the board, and upon no fulfillment, firms were fined \$300,000. The authors (2020) found that announcement returns decreased on average by 1.2%. We contribute to this literature by providing evidence on the impact of the board gender quota on firm value in another context and where the policy implementation details differ. Firstly, the French quota was implemented in a way that allowed firms a relatively longer time to comply with the law. In comparison to Norway’s two-year transition period and California’s slightly over one-year transition period to avoid penalties for non-compliance, French firms were allowed six years to change their boards. Secondly, the consequence of failure to fulfill the quota was relatively softer in France than in Norway. In France, firms were punished with suspension of paying director fees, but in Norway, it was dissolution. Finally, France had a worse female supply setting relative to Norway (Adams and Kirchmaier, 2015). Ultimately, we shed light on what factors play a crucial role in how firm value is affected by the implementation of a new gender quota at the board level.

Furthermore, we contribute to the literature on how board gender quotas affect the characteristics of the board members. Several papers cover how the skills and experience of board members changed as a result of the board gender quota (see Ferreira et al., 2021; Ahern and Dittmar, 2012; Bertrand et al., 2019). In particular, Ferreira et al. (2021) compared the characteristics of the females hired before and after the quota. The authors (2021) found that female directors hired after the quota were more independent, experienced, internationally diverse, and no less academically qualified than those hired before the quota. However, they make no conclusions about how the females hired after the quota differed from their male predecessors. As such, we contribute to the literature by not only comparing the skills of the new and incumbent females but also by comparing the new females to their male predecessors.

2 Data and Methodology

2.1 Data Gathering

We primarily used Refinitiv Eikon to obtain the necessary data for our study. We downloaded a large dataset containing variables for both the event study and the long-term analysis to ensure that the same companies participated in both studies. We limited our dataset to public companies to ensure similar characteristics and also follow the same approach as in Ahern and Dittmar (2012). For the event study, we used a cross-sectional dataset with company names and corresponding variables’ values on the 20th of January in 2010, the main date of our event study. These variables included GICS Sector, GICS Industry Group, board size, board gender diversity, market capitalization, and total assets. Stock returns for a five-day period, two days before until two days after 2010-01-20, were also acquired. We downloaded these statistics for our treatment group, companies incorporated in France listed on the Euronext Paris, and for our control group, companies incorporated in the USA listed on a US exchange.

A panel dataset was used for the long-term study. We gathered data for all companies, both French and US, between 2009 and 2021. The variables we obtained included board size, board gender diversity, market capitalization, annual stock return, common stock, assets, ROE, and board-specific skills. The latter variable is defined as the percentage of board members who have either an industry-specific background or a strong financial background. We gathered all variables for both our treatment and

control groups.

We had to exclude some firms from our study as Refinitiv Eikon did not have data points for all variables between 2009 and 2021 for all firms. The main reason for exclusion was the lack of data on board gender diversity. After screening the raw data, we were left with a sample size of 69 French firms for the event and the long-term study. To ensure a robust result, we collected data on the remaining French firms in the SBF120 index using alternative sources, either excluded due to a few missing data points or having no data at all. CapitalIQ was used to gather financial information, and data on board gender diversity was collected from the companies' annual reports.

The process of acquiring female board participation from annual reports was a systematic one. We first identified the members of the board of directors in the annual report and calculated the board gender diversity by dividing the number of female directors by the total number of board members. Employee representatives were included in our calculations to match with the data downloaded from Refinitiv Eikon. To ensure the accuracy of the board gender diversity, we verified the gender of each person. We first checked if pronouns were mentioned in the description of the director. When no pronouns were used, we visually determined the gender. This involved looking at the attached photograph of the board members. If there was no photography available, we determined the gender based on the first name of the director. However, it can be challenging to distinguish between male and female names, especially for non-French speakers. For instance, François and Françoise are male and female names respectively. Therefore, we consulted the Namsor database, which provided the frequency of males and females in France having the inserted name, to guide us in our decision-making process.

With the downloaded data, Tobin's Q was calculated using the following formula, per Ahern and Dittmar (2012):

$$Tobin'sQ = (TotalAssets - CommonEquity + MarketEquity)/TotalAssets \quad (1)$$

Tobin'sQ is calculated as the sum of *TotalAssets* (reported) and *MarketEquity*, less *CommonEquity* (reported) divided by *TotalAssets*. Market equity, in turn, is the aggregated market value, which is determined by multiplying the price per share by the total number of outstanding shares on Refinitiv Eikon.

Our sample size of French firms used in the event study is 89, and in the long-term study, a total of 93 firms were used, resulting in 1116 firm-year observations.

2.2 Summary Statistics

Table 1: Summary Statistics for French firms

Year	Median Tobin's Q	Mean Board Gender Diversity	Mean Board Tenure	Mean Board-specific Skills	Mean Board Size
2010	1.69	14.74	6.01	36.86	12.98
2011	1.68	19.70	6.27	37.60	12.92
2012	1.70	23.92	6.18	37.68	13.02
2013	1.73	27.95	6.29	35.14	12.99
2014	1.78	31.45	6.45	28.55	12.99
2015	1.74	35.12	6.81	31.48	13.12
2016	1.64	40.13	6.60	30.30	13.13
2017	1.57	43.21	6.44	29.01	13.19
2018	1.53	43.80	6.99	31.08	13.06
2019	1.55	44.90	7.13	31.43	12.90
2020	1.53	44.99	7.02	30.70	13.06
2021	1.50	45.57	7.02	30.58	13.19

Note: Tobin's Q is the only column displaying the median value, as that is what is used to calculate the industry-adjusted Tobin's Q. Board-specific skills is defined as the percentage of board members either possessing industry-relevant skills or a strong financial background.

Table 1 presents the summary statistics for French companies, while Table 2 displays the corresponding statistics for US companies. It is interesting to note that the average board size in both

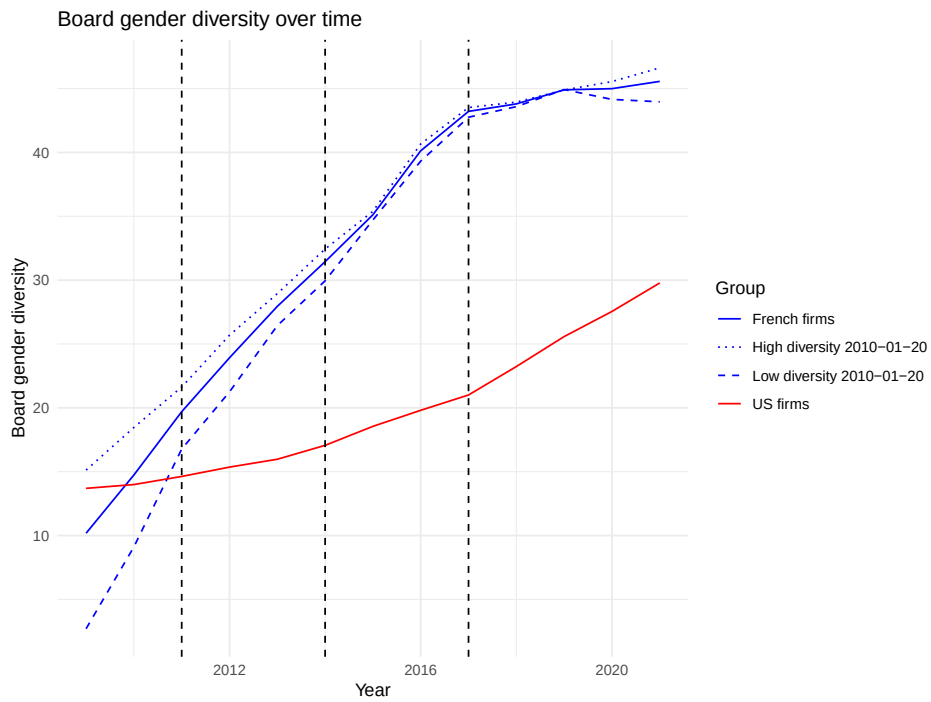


Figure 1:

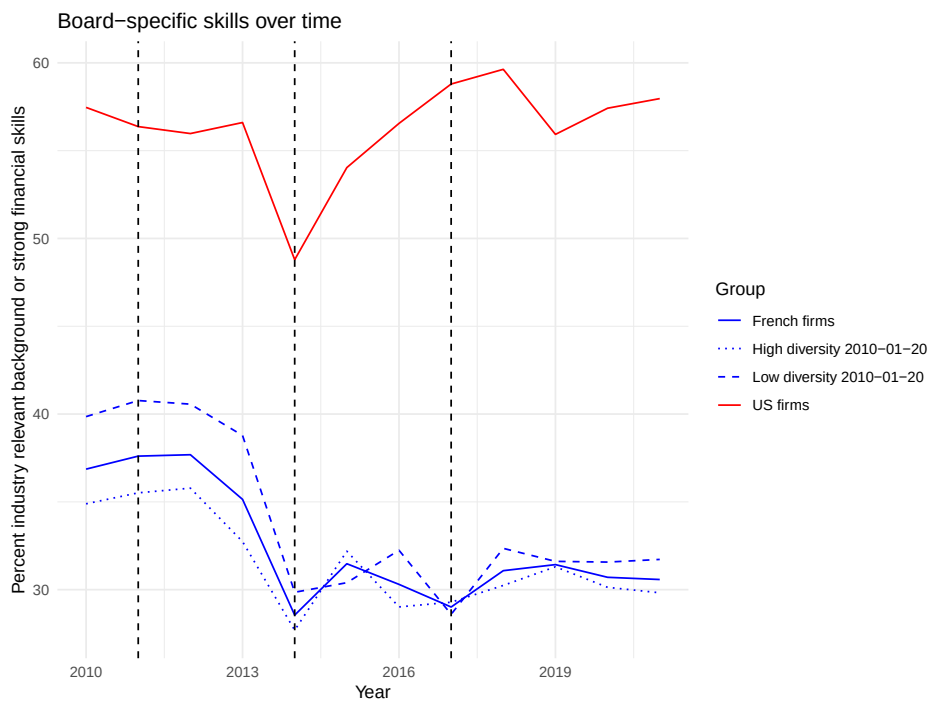


Figure 2:

Table 2: Summary Statistics for US firms

Year	Median Tobins Q	Mean Board Gender Diversity	Mean Board Tenure	Mean Board-specific Skills	Mean Board Size
2010	2.59	13.99	9.43	57.46	10.42
2011	2.47	14.63	9.60	56.37	10.49
2012	2.41	15.36	9.67	55.97	10.51
2013	2.36	15.97	9.84	56.60	10.50
2014	2.30	17.06	9.83	48.79	10.60
2015	2.26	18.55	9.80	54.04	10.60
2016	2.20	19.81	10.56	56.55	10.66
2017	2.16	21.01	10.66	58.80	10.68
2018	2.10	23.23	10.53	59.63	10.67
2019	2.04	25.58	10.55	55.93	10.76
2020	1.97	27.55	10.45	57.42	10.78
2021	1.95	29.78	10.24	57.96	10.96

Note: Tobin's Q is the only column displaying the median value, as that is what is used to calculate the industry-adjusted Tobin's Q. Board-specific skills is defined as the percentage of board members either possessing industry-relevant skills or a strong financial background.

countries does not change much throughout the twelve-year period. In 2010, the average French board size was 12.98, and the same number was 13.19 in 2017 and 2021. This first suggests that the French gender quota most likely had no effect on the board size. Additionally, the average board tenure has increased in France from approximately six years in 2010 to 6.4 years in 2017. This indicates that French firms tended to replace male board members rather than hire additional female directors to comply with the law.

Figure 1 shows that female board participation increases more in France than in the US as a result of the quota. Three vertical dashed lines have been added to the plot. The first significant date is January 2011, when the new law was enacted. The second date is January 2014, when companies were mandated to have a minimum of 20% board gender diversity. The last date is January 2017, when the final requirement of 40% female board participation needed to be met. While France starts at a lower level of 10.18% in 2009 compared to 13.70% in the US, it ultimately ends at a considerably higher level of 45.57% in 2021 compared to 29.78% in the United States. The French increase appears to be moderate, with a slightly steeper curve in the initial years and a flatter curve closer to 2017 and then becoming almost horizontal after the threshold of 40% is reached. The almost straight line in the graph up to 2017 suggests that firms gradually replaced male board members with female candidates instead of drastically hiring women right before the deadline to meet the requirement.

Figure 2 demonstrates that the trends in board skills are consistent in France and the US, with lines following the same pattern. Similar to Figure 1, the same three vertical lines are displayed here. In the US, the percentage of board members who have either an industry-specific background or strong financial background in 2010 was, on average, 57.46%, and in 2021, it is 57.96%. In France, the change is larger between the two years, 36.86% in 2010 and 30.58% in 2021. This means that the value for the US is nearly constant, while there has been a slight decrease in board skills in France. However, when testing, the difference in means is not statistically significant in any of the countries.

Lastly, Table 1 and 2 indicate that Tobin's Q decreased over time in both countries, making it hard to draw a conclusion about how the quota affected French firms. Given the observed decline in Tobin's Q in the United States as well, it is likely attributable to a global trend rather than solely the quota.

2.3 Corporate Governance in France, Europe and the US

Formally adopted on January 27, 2011, the Copé-Zimmerman law requires a minimum of 20% female representation on company boards by January 2014, with the proportion increasing to 40% by January 2017. This quota applies to all board members, except for employee representatives, regardless of whether they are independent or not. Sociétés Anonymes (limited liability companies), Commandites par actions (limited partnerships), and Societas Europaea (European Societies) are all subject to this law. Non-listed companies are also subject to the law, provided they have at least 500 employees or

have earned €50 million over the previous three years. Firms that failed to adhere to the quota were punished with suspension of paying director fees (République Française, 2011). The law was submitted to the French National Assembly on 2009-12-03, and was adopted in the first reading on 2010-01-20. Parliamentary debates continued from that point until it was officially adopted in January 2011. Companies listed on the Euronext Paris that are not based in France are not obligated to comply with the law (Ginglinger and Raskopf, 2023).

The issue of quotas on boards has been a topic of extensive debate in Europe for several years. Norway became the first country to adopt a law in December 2003 which mandated that at least 40% of directors must be female by July 2005. However, voluntary compliance failed, and in 2006, the law became compulsory with a two-year transition period. Firms that failed to adhere to the quota would be dissolved (Ahern and Dittmar, 2012). Additionally, hard laws regarding women on boards have been adopted by several European countries, such as Italy, Spain, the Netherlands, Belgium, Austria, and Germany. Soft laws have been adopted for boards in Luxembourg, the UK, and Sweden (Ferreira et al., 2021). In November 2022, the European Commission adopted a directive that sets a minimum objective that listed companies in Europe must have 40% of the underrepresented gender in non-executive board member positions by 2026 (European Commission, 2022).

During the six-year transition period for French firms between 2011 and 2017, there was no gender quota in place in the US. However, in September 2018, California passed a law that required one female director on the boards of listed companies. The law stated that all companies headquartered in the state must have at least one female director by the end of 2019. Firms that did not fulfill the quota were fined \$300,000. This gender quota was overturned by a Californian court in May 2022. We still determine the United States as a valid control group since only one of the 50 states had a gender quota, and it was implemented after the main period of interest between 2011 and 2017.

While European companies could have been a suitable control group, most of them had already implemented board gender quotas or soft laws during this period. Therefore, the USA was the best country to control with as the largest developed economy with no quota on female directors during the period of interest. This is also in line with the choice of placebo group used by Ahern and Dittmar (2012) in their Norway study and Ginglinger and Raskopf (2023) who looked at how the French gender quota affected E&S performance. Additionally, French and US boards share similar characteristics (Ferreira et al., 2021), further strengthening the choice of control group.

2.4 Methodology

As we categorize the quota’s impact on firm value by looking at it from a short- and long-term perspective to determine the causal relationship, we have divided the methodology into two separate parts. Firstly, we investigate the immediate impact on stock prices by conducting an event study that analyzes the market reaction when the gender quota was first brought up in parliament. Secondly, we address the firm’s long-term value by studying if there is a relation between the forced change in female board participation and the change in firm value (*Tobin’s Q*). The two different methodologies used are further specified below.

2.4.1 Event Study

Although the French parliament passed the statute on the 27th of January, 2011 (République Française, 2011), discussions in parliament started earlier (Ferreira et al., 2021). Therefore, the stock market would have already anticipated the law’s effects before it was formally enacted, making the 27th of January, 2011, an inappropriate time to study. To investigate the market’s actual response, we selected the 20th of January, 2010, as the date of our event study. This was the day when the proposed gender quota was first brought up for discussion in parliament (Ginglinger and Raskopf, 2023). On the same day, an article about the revolutionary quota proposal was France24’s second most-read article released. The wholly government-owned news media France24 called the article ”Shaking up the corporate world”, emphasizing its disruptive nature. In the article, it is brought up how a center-right lawmaker describes the bill as ”a much-needed electric shock” (France 24, 2010) emphasizing the unexpectedness of the law. Thus, 2010-01-20, is a suitable day to study the stock market’s reaction.

Our study was conducted over a period of five consecutive days, two days before the news announcement and two days after, 2010-01-18 to 2010-01-22, in line with Ahern and Dittmar (2012). These dates have also been checked to verify that they occur during weekdays when the stock market is open

for trading. We used industry-adjusted stock returns to control for industry effects. Before controlling for industry effects, it was ensured that firms with higher or lower female board participation did not tend to belong to a particular industry. We also confirmed that every French firm belonged to a GICS industry group with at least five US firms. To calculate the industry-adjusted returns, we subtracted the median stock return of US firms within the same GICS Industry Group, as per the approach of Ahern and Dittmar (2012).

The French firms were divided into two groups based on their female board participation at the time of the event study, in line with the methodology of Ahern and Dittmar (2012). One group is called High diversity firms, and the other is called Low diversity firms. This was to examine whether there was a difference in returns for firms closer or further away from the threshold. We created two approximately equally large groups, one group with a female board participation under or equal to the median of 8.33%, and one group with a board gender diversity above the median. The number of firms in the High diversity group was 41, and 48 companies belonged to the Low diversity group, resulting in a similar proportion between the two groups as Ahern and Dittmar (2012) had.

The Low diversity group only had an average female board participation of 3.68% at the time of the event study, while the High diversity group had an average of 15.57%, meaning that the first group only fulfilled 9.20% of the 40% quota while the second fulfilled 38.93% of the threshold value. This means that the second group needs to replace a much larger quantity of board members compared to the first group. As Table 3 shows, the two groups are not dissimilar when examining several characteristics, making them two good groups to compare. The only significant difference is female board participation, as that variable was used to define the two groups. Table 4 shows that the two groups are also not statistically different in terms of which sector they belong to. The only significant difference is that real estate firms are more likely to have fewer females on their boards. In their study, Ahern and Dittmar (2012) also had one significant sector difference but concluded that as it is only one sector, it does not affect the result’s credibility. For prudence, an additional test is run to ensure there is no correlation between pre-quota board gender diversity and industry belonging. Fisher’s exact test of the difference in the distributions of industries by female board participation finds no statistical difference between the likelihood of a firm to be in any one particular sector based on its percentage of female directors in 2009. This implies that differences between the two groups in subsequent regressions should stem from the difference in board gender diversity.

Table 3: Differences in Characteristics 2010-01-20

Company Characteristics	All firms	Low diversity	High diversity	Difference
Board Size	12.51	12.36	12.68	-0.32
Percent Female Directors	9.22	3.68	15.57	-11.89***
Tobin’s Q	1.49	1.49	1.51	-0.02
log(Assets)	23.23	23.01	23.48	-0.47
log(Market Cap)	22.60	22.46	22.76	-0.30
Employees	68375	59195	78905	-19711
Debt/Equity	0.44	0.43	0.44	-0.00

Note: *p<0.1; **p<0.05; ***p<0.01

The values for Tobin’s Q display the median, all others show the mean

The event study is performed per the methodology of Ahern and Dittmar (2012). First, we calculate the means in industry-adjusted stock return for all French firms and the two subgroups divided based on their board gender diversity on the 20th of January, 2010. We also look at whether these means are significant. To further see how the two subgroups differ, we also calculate the difference in mean industry-adjusted return between Low and High diversity firms and test whether the difference is significant. Then, we perform four regressions with different independent variables and controls on the dependent variable industry-adjusted stock return to see whether board gender diversity is a predictor of the result and how that differs between the two previously mentioned groups. The four main variables of interest are the ones showing the relationship between board gender diversity and industry-adjusted returns. These include board gender diversity, a dummy variable for companies belonging to the subgroup High diversity, a dummy for all French companies interacting with the dummy for High diversity, and the French dummy interacting with board gender diversity. Control

Table 4: Differences in Sectors

GICS Sector	Low diversity	High diversity	Difference
Communication Services	14.58	12.20	2.39
Consumer Discretionary	14.58	14.63	-0.05
Consumer Staples	4.17	9.76	-5.59
Energy	6.25	2.44	3.81
Financials	4.17	12.20	-8.03
Health Care	6.25	4.88	1.37
Industrials	22.92	26.83	-3.91
Information Technology	8.33	4.88	3.46
Materials	4.17	7.32	-3.15
Real Estate	12.50	2.44	10.06*
Utilities	2.08	2.44	-0.36

Note: *p<0.1; **p<0.05; ***p<0.01

The values for Tobin's Q display the median, all others show the mean

variables are board size, company market capitalization, and total assets between log(market cap) and log/assets), similar to Ahern and Dittmar (2012).

2.4.2 Long-term

Our analysis that assesses the long-term consequences of the gender quota on firm valuation uses Tobin's Q as the dependent variable, in line with prior research (Coles et al., 2008; Yermack, 1996; Ahern and Dittmar, 2012). Tobin's Q is the most suitable variable to examine, as it measures the value the company has above its assets, including future growth opportunities. It is calculated in accordance with the formula stated in Equation 1. By studying this measure, we can investigate the company's ability to generate value and whether this ability has changed after the mandated changes of directors resulting from the law. This was also the variable used by Ahern and Dittmar (2012), making our results comparable.

To ensure the validity of our findings, we have taken measures to address any potential endogeneity concerns. One issue is that companies may respond to the gender quota in different ways. Another issue is that they may attribute their positive or negative performance to the quota, even when it is not related. For example, a company may hire more women before experiencing favorable(unfavorable) financial outcomes and attribute the success(failure) to the gender quota. We have accounted for these possibilities in our analysis. To overcome these problems, we adopt an instrumental variable approach, taking inspiration from studies by Ahern and Dittmar (2012) and Stevenson (2010). Specifically, we leverage the pre-quota cross-sectional variation in board gender diversity as an instrument to capture exogenous variation in mandated changes in female board participation over time.

Ahern and Dittmar (2012) used data from one year before the implementation of the Norwegian gender quota. In our study, we have utilized data from 2009, which is two years before the French law was implemented in 2011. We have chosen this year to further ensure instrument exogeneity, meaning no correlation between the instrument (z_i) and the error term (u_i), ($corr(z_i, u_i) = 0$). Going further back would eliminate many companies from the study due to data unavailability. For the firms where the data on board gender diversity is available, we compared the 2008 and 2009 values, concluding that they were very similar, indicating that the board composition in 2009 was not impacted by the law in 2008.

Since the gender composition of a company's board is not random but something that is actively chosen, we observe whether there are any links between the pre-quota board gender diversity and certain firm characteristics and sectors. As shown by Table 3 and Table 4, no such relationship is found apart from in one sector, similar to Ahern and Dittmar (2012). To further ensure that there are no links between industries and pre-quota cross-sectional variation, Fisher's exact test finds no significant difference ($p = 0.78$) in industry distribution based on pre-quota board gender diversity. In this analysis, we also adjust the dependent variable for US industry returns as a precautionary measure, per Ahern and Dittmar (2012). Hence, the dependent variable is not Tobin's Q but industry-adjusted

Tobin’s Q. Similar to the event study, we subtract the median value of American firms in the same GICS Industry Group.

To identify the effect of the gender quota on firm value, we estimate the following equation:

$$Q_{i,t} = \alpha + \beta \text{percentfemaledirectors}_{i,t} + \theta_i + \pi_t + \epsilon_{i,t} \quad (2)$$

In this analysis, we consider the industry-adjusted Tobin’s Q for firm i during the year t , denoted as $Q_{i,t}$. The model has a constant, α , and a coefficient β that establishes the strength of the relationship between $Q_{i,t}$ and the *percentfemaledirectors*. To control for any constant differences in $Q_{i,t}$ among different firms, we add firm fixed effects (denoted as θ). In all regressions with firm-fixed effects, we cluster standard errors within the company. This is to avoid potential problems of serial correlation in the time series of within-firm variation that can be observed in differences-in-differences variables (Bertrand et al., 2004). In addition, we add time-fixed effects (denoted as π) to account for any booms or slumps during particular years. Furthermore, we run placebo tests for US firms alongside our regressions on French firms. We instrument the percentage of female directors using female board participation in 2009 interacted with year dummies.

In order to ensure the robustness of our analysis, we conducted a similar instrumental variable regression using a different performance measure, return on equity (ROE). ROE is calculated as profit before tax divided by book equity, and it reveals a company’s profitability by measuring how much profit it generates with the money the shareholders have invested. This measure is essential for our study because it assesses a company’s efficiency without considering market value, as Tobin’s Q does. A high ROE value indicates a profitable and efficient company that generates a good return on the capital invested by the company’s shareholders.

Another important part of our long-term study is to see how the skill level of the board is impacted by the mandated quota. This is interesting because of two reasons. Firstly, Ahern and Dittmar (2012) saw a decline in the skill level of the board as incumbent board members were ousted in favor of female directors who could help fulfill the quota. Secondly, the authors attribute the decline in firm value to the decline in board skills (Ahern and Dittmar, 2012). Hence, this study will examine how the Copé Zimmerman law affected the skill level of the board. Our dependent variable is board-specific skills, which is defined as the percentage of board members who either have an industry-relevant background or strong financial skills. As boards provide value by monitoring and advising the CEO (Fama and Jensen, 1983), we posit that these two traits serve as good indicators of expertise.

We will investigate how board-specific skills is impacted by the increased board gender diversity stemming from the French gender quota per the instrumental variable methodology used by Ahern and Dittmar (2012) to study how education and experience among board members were affected by the Norwegian law.

We thus use the following equation:

$$\text{boardspecificskills}_{i,t} = \alpha + \beta \text{percentfemaledirectors}_{i,t} + \theta_i + \pi_t + \epsilon_{i,t} \quad (3)$$

We also aim to examine the effects of the law on board size and board tenure. Our goal is to assess whether new female directors were added following the law or if they replaced existing male directors. Once more, we employ an instrumental variable method, using the pre-quota percentage of female board participation to instrument for exogenous changes to corporate boards.

3 Results

3.1 Event Study

Table 5 displays the mean industry-adjusted return in percent during the five-day period mentioned earlier for all 89 firms, High diversity firms, and Low diversity firms, along with the difference in means between both subgroups that were divided based on their female board participation on 2010-01-20. The table reveals that the mean for all firms was positively significant, with a value of 0.76%. The mean value for Low diversity firms was also positively significant, with a slightly higher value of 1.14%, in contrast to the insignificant result for diversity firms, which was also positive at a magnitude of

Table 5: Mean industry-adjusted stock returns (%)				
Statistic	All firms (1)	Low diversity (2)	High diversity (3)	Difference (2) - (3)
Mean	0.76** (3.35)	1.14** (2.85)	0.31 (3.85)	0.83 (0.73)
Observations	89	48	41	

Note:

*p<0.1; **p<0.05; ***p<0.01

Columns two and three display the results for the two French subgroups

0.31%. The difference in means was not statistically significant at a value of 0.83%, indicating that the results for both subgroups did not differ significantly.

Table 6 provides further insight into the stock market’s reaction to a news announcement. The table presents the results of four different regressions, each with the dependent variable industry-adjusted stock return. Independent variables include different combinations of High diversity Dummy, Board Gender Diversity, French Dummy, Board Size, and log(Assets), all of which are crucial in understanding the complex dynamics of the stock market.

The most interesting variables for our study are the four at the top displaying the relationship between female board participation and industry-adjusted returns: High diversity dummy, French Dummy interacted with High diversity Dummy, Board gender diversity, and French Dummy interacted with Board gender diversity. None of these variables show a significant result, meaning there was no significant relationship between female board participation and industry-adjusted return at the time of our event study when the quota proposal was first brought up in parliament and sent shock waves throughout France (France 24, 2010).

Table 6: OLS regression on industry-adjusted announcement returns (%)

	<i>Dependent variable:</i>			
	(1)	(2)	(3)	(4)
Board gender diversity		0.001 (0.015)		0.015 (0.015)
High diversity dummy	-0.164 (0.323)		0.024 (0.321)	
French dummy:High diversity dummy	-0.529 (0.786)		-0.499 (0.771)	
French dummy:Board gender diversity		-0.042 (0.046)		-0.033 (0.045)
French dummy	0.900 (0.563)	1.087* (0.591)	0.914 (0.564)	1.063* (0.590)
Board size			0.107** (0.053)	0.102* (0.053)
log(Assets)			-0.537*** (0.101)	-0.551*** (0.101)
Constant	0.106 (0.279)	-0.030 (0.248)	11.133*** (2.068)	11.307*** (2.081)
Observations	658	658	658	658
R ²	0.007	0.006	0.049	0.050
Adjusted R ²	0.002	0.002	0.042	0.043
Residual Std. Error	3.355 (df = 654)	3.356 (df = 654)	3.288 (df = 652)	3.286 (df = 652)
F Statistic	1.501 (df = 3; 654)	1.396 (df = 3; 654)	6.759*** (df = 5; 652)	6.916*** (df = 5; 652)

Note: *p<0.1; **p<0.05; ***p<0.01

3.2 The Impact of the Quota on Tobin's Q

This section analyzes the impact the exogenous quota had on the firms' Tobin's Q, both during and after the implementation period. To identify the effect of the gender quota on firm value, we use the previously described instrumental variable approach and Equation 2.

Table 7: Instrumental variables regression on Industry-adjusted Tobin's Q

	<i>Dependent variable:</i>	
	Industry-adjusted Tobin's Q (France)	(USA)
Board gender diversity	-0.279 (1.160)	-0.163083 (0.654514)
Time fixed effects	Yes	Yes
Firm fixed effects	Yes	Yes
Observations	1,116	7,356
R ²	0.727	0.7796
Adjusted R ²	0.700	0.7592
Residual Std. Error (df = 1012, 6732)	2.299	4.65

Note:

*p<0.1; **p<0.05; ***p<0.01

Table 7 displays the relationship between the forced gender diversity on the boards and Tobin's Q for both French and US companies. The instrumental variable regression accounts for both firm and time-fixed effects. The findings reveal that the board gender quota did not significantly influence Tobin's Q for French companies. The coefficient shows a negative result of roughly -0.28%, but it is statistically insignificant. Similarly, for US firms, the outcome also shows a negative coefficient of roughly -0.16%, but it is also not statistically significant. The non-significant result is expected because US firms serve as a control group.

Table 9 (see Appendix) demonstrates the reduced form regressions, with board gender diversity as the dependent variable and the independent variables being year dummy variables interacting with the 2009 female board participation for the firms. The result shows that the interaction variables are insignificant for all the years, indicating that there is no year in which board gender diversity had a significant impact on Tobin's Q. This implies that even though the board gender diversity increased over time, it did not impact the firm's value significantly at any point in time.

Table 8 illustrates the instrumental variable regression on board-specific skills, defined as the percentage of board members having either a strong financial background or industry-relevant skills, in relation to the board gender diversity quota. The model for the regression is previously described in Equation 3. The result from the instrumental variable regression shows a positive coefficient of 0.16%, however insignificant. This means that the mandated board changes did, in fact, not decrease board skills, as the regression result shows a positive although insignificant result.

The result of the instrumental variable regression in Table 13 (see Appendix) shows that the board size cannot be confirmed to have changed as a result of the law. The point estimate for board gender diversity was -0.208 and insignificant. Similarly, the instrumental variable coefficient for board gender diversity in Table 14 (see Appendix) was 0.040 for the regression on board tenure, indicating that the law did not have a significant impact on it.

Table 8: Instrumental variables regression on board-specific skills

	<i>Dependent variable:</i>
	Board-specific skills
Board gender diversity	0.161 (6.148)
Time fixed effects	Yes
Firm fixed effects	Yes
Observations	1,116
R ²	0.542
Adjusted R ²	0.496
Residual Std. Error	12.713 (df = 1013)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

4 Discussion

The short- and long-term studies in the result section are consistent regarding the board gender quota’s impact on the firm value of the affected French companies. In the short term, Table 5 and 6 provide insight into the stock market’s reaction to the news announcement about the board gender quota. Table 5 shows the difference between the average industry-adjusted returns for the High diversity group and Low diversity group to be 0.83%. However, the result is insignificant. Moreover, the same study done on our control group of US firms showed no significant result because they were not affected by the announcement of the French board gender quota. In contrast, Ahern and Dittmar (2012) found that Low board gender diversity firms had, on average, a 3.56% worse return than the High diversity group in relation to the announcement of the board gender quota. The authors’ (2012) findings were significant. Also, the stock market reaction for Californian firms in response to the quota announcement was -1.2% on average return (Greene et al., 2020). Thus, we can see that for the French law, the stock market reaction differed in that it was of a smaller magnitude and insignificant, compared to the reactions to the Californian and Norwegian gender quotas.

Furthermore, the OLS regression in Table 6 shows a similar result. The fact that no one out of the four top variables in the table was significant implies that there was no significant relationship between firms’ industry-adjusted stock return in response to the quota announcement and their female board participation. This goes for both groups of board gender diversity of French firms. In the Norway study, the insignificant variables were the same apart for the country dummy variable interacted with the High diversity dummy variable (Ahern and Dittmar, 2012). This value was 3.5% and significant, implying that the High diversity firms performed relatively better than the Low diversity firms. The corresponding number in our study was -0.5% and insignificant. Hence, a difference in performance cannot be concluded.

Furtado and Rozeff (1987) state that the wealth effects of board shifts reflect the capital market’s view on the change. We argue that the differing stock market reactions to the French and Norwegian (Ahern and Dittmar, 2012) laws are related to investors interpreting the impact of the quota differently in the two scenarios. As such, we will next analyze what could have caused the disparate views by comparing the differences in the long-term firm value development.

The mandated increase in board gender diversity from the French board gender quota does not have a significant impact on the long-term development of Tobin’s Q. Table 7 illustrates that in our instrumental variable regression, using the 2009 pre-quota cross-sectional variation in board gender diversity as our instrumental variable, French firms’ Tobin’s Q decreased by 0.28% on average as a result of the law. However, the result is insignificant. This implies that even though the board gender quota forced an involuntary increase in board gender diversity, it had no significant impact on the value of firms. This is further confirmed by our robustness test using ROE as our dependent variable in Table 12 (see Appendix), where the result was also insignificant. As expected, the result for our

control group, US firms, was insignificant as the firms were not subject to the French law. Our results differ from the paper by Ahern and Dittmar (2012), who found that firm value decreased by 1.9% as a result of the quota. Once again, their result from a different setting was of a larger magnitude and significant. In addition, we also studied whether there was a lag effect by looking at data until three years after 2017. It could be the case that it takes some time before any impact from a change in the board of directors is reflected in the company’s operations and value. Table 9 (see Appendix) shows that at no point in time, the pre-quota variation in board gender diversity interacting with Tobin’s Q variable is significant. These findings are in line with Ahern and Dittmar (2012), who neither found any lagged impact on firm value, although only looking at one year after the quota deadline. Consequently, our result does not show support for the forced board gender diversity impacting firm value under or after the transition period.

Our findings suggest that the impact of a board gender quota on firm value varies depending on the context and is not always negative. Ahern and Dittmar (2012) demonstrate that firm value declines in both the short- and the long-term in their examination of the Norwegian law. Both quotas have had the same overarching goal in society, to increase gender diversity in corporate boardrooms, which they have been successful in achieving. However, one example led to shareholders bearing a cost for the increased female board participation, and not the other. The contrasting results imply that French firms could cope with the transition more successfully than firms in Norway.

A possible explanation as to why no long-term performance effect was found in our French experience could be explained by the law not having a negative impact on board-specific skills. Ahern and Dittmar (2012) considered the decline in board skills to have caused the decline in firm value. In Table 8, our instrumental variable regression on board-specific skills, which is defined as the percentage of board members having either an industry-relevant background or strong financial skills, with board gender diversity as the independent variable, illustrates an insignificant relationship between the two. Moreover, as can be seen in Table 13 and Table 14 (see Appendix), the board size nor average board tenure was not significantly affected by the forced increased board gender diversity. This suggests that board size remained roughly constant and that incumbent male directors were replaced instead of additional females hired. Thus, our result implies that French firms, in contrast to Norwegian companies, were able to replace male directors with female board members without negatively impacting the board-specific skill level. Ferreira (2021), who also studied the effects of the French board gender quota, found that females who entered the board as a result of the quota were more independent, experienced, and equally academically educated compared to the females hired before the quota. Our result thus adds to this finding as it shows that even though there was a forced replacement of board members, the new females in France were not only as skilled as the incumbent female board members as found by Ferreira et al. (2021), but equally as skilled as their male predecessors. Ahern and Dittmar (2012) found that the new female directors who entered the Norwegian firms’ boards had significantly less experience and relevant characteristics than the previous male directors. The authors (2012) suggest that the quota posed a demand shock for women with relevant board characteristics, and the supply did not match, causing subsequent performance deterioration. However, Adams and Kirchmaier (2015) show that Norway had a better female supply setting than France. The authors (2015) examine what country-specific factors pose challenges for women to enter the boardroom during the Norwegian transition period (2012) and one year prior to our French event study. Adams and Kirchmaier (2016) conclude that France had a lower level of female labor supply is less family-friendly, which reduces women’s ability to manage both family and work and has a culture of strong family values, making it difficult for women to accept board positions, relative to Norway. As such, this suggests that Norwegian firms should have had less of a problem finding suitable females relative to French firms. Hence, the supply factor fails to explain why French firms were more successful than Norwegian firms (Ahern and Dittmar, 2012) in maintaining their skill level on the board.

We argue that if firms had to rush to fulfill the quota, it affected their ability to maintain the same skill level on the board. In Norway (Ahern and Dittmar, 2012), board gender diversity rapidly increased between 2006 and 2007, relative to the previous years when the quota was voluntary. This is illustrated by the spike during those years in Figure 3. The authors (2012) also find that their significant result on Tobin’s Q is driven by the two years before the quota deadline. In our study, we find that France was able to adjust smoothly to the new legislation. In Figure 1, board gender diversity can be seen increasing with a roughly constant slope between the years 2011 and 2017. The result is similar for both the High and Low diversity groups. This suggests that firms were not required to



Figure 3: Average gender diversity between 2001-2009 in Norway (Ahern and Dittmar, 2012)

make drastic adjustments at any point in time. Evidently, the various settings pushed Norwegian firms to behave differently than French firms. Next, we explain potential reasons why these firm behaviors could be seen.

The reason for the different adjustment patterns can be explained by the various context and policy implementation details in the quota. Firstly, the sanction upon failure to comply with the quota was more extensive in Norway (Ahern and Dittmar, 2012) relative to France. In the case of France, firms were punished with suspension of paying director fees if they did not fulfill the quota when compliance was due. On the other hand, Norway had a more dramatic outcome upon non-compliance. The company would have been dissolved if the Norwegian firms had not fulfilled the quota in time. As such, Norwegian firms were facing a more acute threat. This can explain why Norwegian firms' board gender diversity spiked as the deadline approached (Ahern and Dittmar, 2012). Secondly, the time window firms were allowed to adapt to the quota is another critical explanation for the various behaviors. Bagues and Campa (2021) emphasized that in the enactment of a gender quota, more time allowed for adaptation is needed to find relevant and desired candidates. In France, firms were allowed six years to change their boards, compared to Norwegian firms, which had two years. The different lengths of the transition periods likely influenced the way firms dealt with the issue. French firms were probably less pressured to conform to the quota and had more time to find a suitable replacement plan. Consequently, the process of substituting the male predecessors with equally skilled females was not inhibited by a relatively worse supply. In contrast, Norwegian firms faced harder punishments and had less time. As a result, they probably rushed the process and were unsuccessful in finding female board candidates with similar skill levels (Ahern and Dittmar, 2012), despite having a relatively better supply setting.

Ultimately, our result sheds light on the critical factors that are needed for legislators to successfully implement board gender quotas. The evidence supports the idea that board gender quotas do not necessarily have a negative impact on firm value, which shows that the goal of increased board gender diversity is attainable without shareholders bearing a financial cost. By comparing results, policy implementation details, and context with the Norwegian study (Ahern and Dittmar, 2012), we can identify the factors that made the French experience more successful. In our study, we could see that firms were effective in maintaining their board-specific skill level despite replacing their freely selected board members, as indicated by the board size and tenure not being affected by the law. This evidence can help alleviate potential concerns about board gender quotas in the corporate sector and address worries regarding the soon-to-be-implemented EU directive on board gender diversity. The ability to keep the board-specific skill level relatively unchanged is what we suggest enables French firms to not deteriorate in firm value. This then can be explained by the specifics of the French gender quota. The characteristics of the Copé-Zimmerman law allowed firms a total of six years and had a relatively

lenient sanction upon failure to fulfill the quota. We argue that it was the uniqueness of the time period and possibly the penalty level that led to French firms' ability to *gradually* adapt to the new quota and not make drastic changes at the expense of firm performance. Policymakers can draw lessons from these findings to successfully achieve increased gender diversity in corporate boardrooms without incurring a financial cost.

5 Conclusion

In conclusion, we find no support for the French board gender quota impacting firm value negatively. Although the law forced an involuntary increase of female board members to meet the threshold stipulated by the quota, we found no significant results in either the short-term study on immediate stock returns at the time of the news announcement or the long-term study on Tobin's Q and return on equity. We found that board size remained roughly constant and was not increased due to the law. Board tenure was also not significantly impacted by the quota. This likely points to the new female directors replacing incumbent male board members instead of being hired as complementary directors. Our results also show no significant signs of the average skill level of boards decreasing.

This study uses methodology from Ahern and Dittmar (2012) who studied the Norwegian gender quota, and compares results. The authors found the quota to affect firm value negatively and believed this to stem from the fact that the quota led to a decline in the average skill level of the boards. These results were driven by the two years before the quota threshold when firms made drastic adjustments to fulfill the quota. We argue that our results showed no adverse effects of the French gender quota as firms did not have to rush to fulfill the quota but could adjust smoothly to the new legislation, as shown by the data. In contrast to the setting for Norwegian firms, which had two years to fulfill the mandated quota, French firms had six years. Also, the French board gender quota had a softer punishment for compliance failure than in Norway, which can reduce the risk of drastic adjustments. This study thus shows that, contrary to prior research, a gender quota does not need to negatively affect firm value, which may provide policymakers guidance on maintaining firm value while providing greater gender equality in the boardroom.

Appendix A: Additional Results

Table 9: Reduced-form regressions on Industry-adjusted Tobin's Q

	<i>Dependent variable:</i>	
	Industry-adjusted Tobin's Q	
	(France)	(USA)
2010:Pct_2009	0.0004 (0.019)	0.060*** (0.016)
2011:Pct_2009	0.005 (0.019)	0.052*** (0.016)
2012:Pct_2009	0.005 (0.019)	0.043*** (0.016)
2013:Pct_2009	-0.0003 (0.019)	0.036** (0.016)
2014:Pct_2009	0.009 (0.019)	0.035** (0.016)
2015:Pct_2009	0.005 (0.019)	0.033** (0.016)
2016:Pct_2009	0.001 (0.019)	0.026 (0.016)
2017:Pct_2009	-0.005 (0.019)	0.022 (0.016)
2018:Pct_2009	-0.003 (0.019)	0.015 (0.016)
2019:Pct_2009	0.001 (0.019)	0.009 (0.016)
2020:Pct_2009	0.004 (0.019)	0.007 (0.016)
Time Fixed Effects	Yes	Yes
Firm Fixed Effects	Yes	Yes
Observations	1,116	7,356
R ²	0.001	0.004
Adjusted R ²	-0.101	-0.088
F Statistic	0.081 (df = 11; 1012)	2.675*** (df = 11; 6732)

Note:

*p<0.1; **p<0.05; ***p<0.01

Table 10: Reduced form regressions (Board specific skills)

	<i>Dependent variable:</i>
	Board-specific skills
2011:IV	−0.104 (0.218)
2012:IV	−0.052 (0.218)
2013:IV	−0.025 (0.218)
2014:IV	0.144 (0.218)
2015:IV	0.350 (0.218)
2016:IV	0.121 (0.218)
2017:IV	0.294 (0.218)
2018:IV	0.234 (0.218)
2019:IV	0.301 (0.218)
2020:IV	0.298 (0.218)
2021:IV	0.273 (0.218)
Time fixed effects	Yes
Firm fixed effects	Yes
Observations	1,116
R ²	0.077
Adjusted R ²	−0.027
F Statistic	3.634*** (df = 23; 1002)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

Table 11: First stage regressions on Board gender diversity

	<i>Dependent variable:</i>	
	Board gender diversity	
	(France)	(USA)
2011	7.626*** (1.276)	1.334** (0.562)
2012	12.057*** (1.276)	2.690*** (0.562)
2013	16.769*** (1.276)	4.216*** (0.562)
2014	20.792*** (1.276)	6.267*** (0.562)
2015	24.914*** (1.276)	9.407*** (0.562)
2016	31.289*** (1.276)	11.061*** (0.562)
2017	35.101*** (1.276)	12.997*** (0.562)
2018	35.857*** (1.276)	16.284*** (0.562)
2019	36.889*** (1.276)	19.417*** (0.562)
2020	36.474*** (1.276)	21.867*** (0.562)
2021	36.153*** (1.276)	24.182*** (0.562)
2011:Pct_2009	-0.262*** (0.096)	-0.051 (0.034)
2012:Pct_2009	-0.283*** (0.096)	-0.097*** (0.034)
2013:Pct_2009	-0.350*** (0.096)	-0.163*** (0.034)
2014:Pct_2009	-0.402*** (0.096)	-0.233*** (0.034)
2015:Pct_2009	-0.446*** (0.096)	-0.354*** (0.034)
2016:Pct_2009	-0.580*** (0.096)	-0.383*** (0.034)
2017:Pct_2009	-0.651*** (0.096)	-0.437*** (0.034)

Table 11: (continued)

2018:Pct_2009	−0.668*** (0.096)	−0.514*** (0.034)
2019:Pct_2009	−0.661*** (0.096)	−0.572*** (0.034)
2020:Pct_2009	−0.611*** (0.096)	−0.607*** (0.034)
2021:Pct_2009	−0.524*** (0.096)	−0.613*** (0.034)
Time Fixed Effects	Yes	Yes
Firm Fixed Effects	Yes	Yes
Observations	1,116	7,356
R ²	0.801	0.509
Adjusted R ²	0.778	0.463
F Statistic	182.849*** (df = 22; 1001)	317.013*** (df = 22; 6721)

Note:

*p<0.1; **p<0.05; ***p<0.01

Table 12: Instrumental variables regression for robustness on ROE

	<i>Dependent variable:</i>	
	Industry-adjusted annual ROE (France)	(USA)
Board gender diversity	−0.018 (0.080)	0.0779636 (0.1674163)
Time Fixed Effects	Yes	Yes
Firm Fixed Effects	Yes	Yes
Observations	816	7356
R ²	−0.084	−0.02736
Adjusted R ²	−0.198	−0.1224
Residual Std. Error (df = 737, 6732)	0.158	1.189
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01	

Table 13: Instrumental variables regression on Board size

	<i>Dependent variable:</i>
	Board size
Board gender diversity	−0.208 (0.983)
Time fixed effects	Yes
Firm fixed effects	Yes
Observations	1,116
R ²	0.667
Adjusted R ²	0.633
Residual Std. Error	2.033 (df = 1013)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

Table 14: Instrumental variables regression on Board tenure

	<i>Dependent variable:</i>
	Average board tenure
Board gender diversity	0.040 (1.060)
Time fixed effects	Yes
Firm fixed effects	Yes
Observations	1,116
R ²	0.635
Adjusted R ²	0.598
Residual Std. Error	2.191 (df = 1013)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

Appendix B: AI disclosure

We have utilized both Gemini and ChatGPT to assist us in the coding process. The AItools have been used to spot coding errors and propose different methods for completing programming operations.

References

- R.B. Adams and D. Ferreira. Women in the boardroom and their impact on governance and performance. *Journal of Financial Economics*, 94(2):291–309, 2009. doi: 10.1016/j.jfineco.2008.10.007.
- Renée B Adams and Tom Kirchmaier. Barriers to Boardrooms. Technical report, 2015. URL <http://ssrn.com/abstract=2192918>http://ssrn.com/abstract_id=2192918www.ecgi.org/wp.
- Renée B. Adams and Tom Kirchmaier. Women on Boards in Finance and STEM Industries. *American Economic Review*, 106(5):277–281, 5 2016. ISSN 0002-8282. doi: 10.1257/aer.p20161034.
- Kenneth R. Ahern and Amy K. Dittmar. The changing of the boards: The impact on firm valuation of mandated female board representation. *Quarterly Journal of Economics*, 127(1):137–197, 2 2012. ISSN 15314650. doi: 10.1093/qje/qjr049.
- Manuel Bagues and Pamela Campa. Can gender quotas in candidate lists empower women? Evidence from a regression discontinuity design. *Journal of Public Economics*, 194:104315, 2 2021. ISSN 0047-2727. doi: 10.1016/J.JPUBECO.2020.104315.
- G. Bernile, V. Bhagwat, and S. Yonker. Board diversity, firm risk, and corporate policies. *Journal of Financial Economics*, 127(3):588–612, 2018. doi: 10.1016/j.jfineco.2017.12.009.
- M. Bertrand, E. Duflo, and S. Mullainathan. How Much Should We Trust Differences-In-Differences Estimates? *The Quarterly Journal of Economics*, 119(1):249–275, 2 2004. ISSN 0033-5533. doi: 10.1162/003355304772839588.
- Marianne Bertrand, Sandra E. Black, Sissel Jensen, and Adriana Lleras-Muney. Breaking the Glass Ceiling? The Effect of Board Quotas on Female Labour Market Outcomes in Norway. *Review of Economic Studies*, 86(1):191–239, 1 2019. ISSN 1467937X. doi: 10.1093/restud/rdy032.
- Norma Burrow, Alexandra Fedorets, and Anna Gibert. The effects of a gender quota on the board of German largest corporations. Technical report, 2018. URL <http://www.deutschlandfunk.de>.
- Jeffrey L. Coles, Naveen D. Daniel, and Lalitha Naveen. Boards: Does one size fit all? *Journal of Financial Economics*, 87(2):329–356, 2 2008. ISSN 0304-405X. doi: 10.1016/J.JFINECO.2006.08.008.
- European Commission. New gender balance targets on company boards, 2022. URL https://ec.europa.eu/commission/presscorner/detail/da/statement_22_7074.
- Eugene F Fama and Michael C Jensen. Separation of Ownership and Control. Technical Report 2, 1983.
- Daniel Ferreira, Edith Ginglinger, Université Paris-Dauphine, and Yasmine Skalli. Marie-Aude Laguna. Technical report, 2021. URL http://ssrn.com/abstract_id=2992213[www.ecgi.global/content/working-papersClosingtheGap:BoardGenderQuotasandHiringPracticesElectroniccopyavailableat:https://ssrn.com/abstract=2992213](http://ssrn.com/abstract_id=2992213www.ecgi.global/content/working-papersClosingtheGap:BoardGenderQuotasandHiringPracticesElectroniccopyavailableat:https://ssrn.com/abstract=2992213).
- France 24. Shaking up the corporate world - Focus, 2010. URL <https://www.france24.com/en/20100120-shaking-corporate-world>.
- E.P.H. Furtado and M.S. Rozeff. The wealth effects of company initiated management changes. *Journal of Financial Economics*, 18(1):147–160, 1987. doi: 10.1016/0304-405X(87)90065-1.
- Edith Ginglinger and Caroline Raskopf. Women directors and E&S performance: Evidence from board gender quotas. *Journal of Corporate Finance*, 83:102496, 12 2023. ISSN 0929-1199. doi: 10.1016/J.JCORPFIN.2023.102496.
- D. Greene, V.J. Intintoli, and K.M. Kahle. Do board gender quotas affect firm value? Evidence from California Senate Bill No. 826. *Journal of Corporate Finance*, 60, 2020. doi: 10.1016/j.jcorpfin.2019.101526.

Tanakorn Likitapiwat, Sirimon Treepongkaruna, Pornsit Jiraporn, and Ali Uyar. Corporate culture, innovation, and female board representation: Evidence from earnings conference calls. *SSRN Electronic Journal*, 2022. ISSN 1556-5068. doi: 10.2139/ssrn.4192014.

République Française. LOI n° 2011-103 du 27 janvier 2011 relative à la représentation équilibrée des femmes et des hommes au sein des conseils d'administration et de surveillance et à l'égalité professionnelle (1) - Légifrance, 2011. URL <https://www.legifrance.gouv.fr/jorf/id/JORFTEXT000023487662/>.

Betsey Stevenson. BEYOND THE CLASSROOM: USING TITLE IX TO MEASURE THE RETURN TO HIGH SCHOOL SPORTS. Technical Report 2, 2010. URL <https://about.jstor.org/terms>.

David Yermack. Higher market valuation of companies with a small board of directors. *Journal of Financial Economics*, 40(2):185–211, 2 1996. ISSN 0304-405X. doi: 10.1016/0304-405X(95)00844-5.